A Regime-Switching Dynamic Reserve Model Linking Relative Dose Intensity and Functional Recovery During First-Line Diffuse Large B-Cell Lymphoma Therapy

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Abstract

Relative dose intensity (RDI) summarizes treatment delivery in diffuse large B-cell lymphoma (DLBCL), but it cannot show when declining reserve changes a patient from treatment-tolerant to treatment-limited. We developed a theoretical regime-switching model linking lymphoma burden, hematologic reserve, functional reserve, comorbidity pressure, and delivered RDI across first-line therapy. Four normalized states represented lymphoma burden $L(t)$, hematologic reserve H(t), functional reserve $F(t)$, and cumulative delivered intensity $D(t)$. A Dynamic Treatment Reserve Index, $DTRI(t) = \frac{\sqrt{H(t)F(t)}}{(1 + 0.25L(t) + 0.25C)}$, governed preserved-reserve, compensated-vulnerability, and treatment-intolerance regimes with illustrative dose multipliers of 1.00, 0.75, and 0.55. Literature informed variable selection and plausible ranges; no patient-level fitting was performed. We examined existence, uniqueness, positivity, boundedness, regime-specific equilibria, Jacobian stability, switching boundaries, and basin behavior. Six-cycle Runge-Kutta simulations, two-dimensional continuation, 500-set Latin hypercube sampling, PRCC, stochastic parameter perturbation, and solver-tolerance checks assessed robustness. In the reference scenario, modeled RDI was 0.674, lymphoma burden fell from 0.750 to 0.149, while DTRI moved from 0.549 to 0.397. High initial reserve increased RDI to 0.785 and reduced final burden to 0.103. Low reserve reduced RDI to 0.553 and left burden at 0.211, severe constraint remained treatment-intolerant for 100.0% of simulated time. Continuation located an illustrative $RDI \geq 0.70$ boundary near equal baseline reserves of 0.745. The 90% uncertainty interval was 0.567-0.753 for RDI and 0.081-0.288 for final burden. Dominant RDI determinants included baseline hematologic reserve (PRCC 0.859), functional reserve (0.872), hematologic toxicity (-0.653), functional toxicity (-0.694), and comorbidity pressure (-0.769). Parameter refinement and biological calibration are ongoing. This mechanistic proof-of-concept provides a mathematically testable framework for locating treatment-tolerance windows that conventional association analyses may not reveal. Its intended uses are serial-variable selection, dose-delivery hypothesis development, and retrospective or prospective validation with lymphoma teams. The expected benefit is earlier recognition of reserve collapse; its research advantages are transparent transitions, reproducible counterfactuals, explicit uncertainty, and separation of simulated thresholds from clinical recommendations.

Keywords: Diffuse Large B-Cell Lymphoma, Relative Dose Intensity, Regime Switching, Dynamic Reserve, Treatment Tolerance, Uncertainty Quantification

Introduction

Diffuse large B-cell lymphoma (DLBCL) is commonly treated with curative-intent immunochemotherapy, yet the ability to maintain treatment intensity varies substantially among patients, particularly with increasing age, comorbidity, and vulnerability. Relative dose intensity (RDI) provides a quantitative summary of delivered treatment relative to the intended regimen, and previous studies have examined its association with outcomes in patients receiving R-CHOP-based therapy **[1-4]**. Dose attenuation, however, is closely intertwined with the patient's underlying capacity to tolerate repeated treatment, making physiological reserve an important consideration when interpreting treatment delivery **[2-5]**. RDI is inherently cumulative and therefore does not fully describe the longitudinal pathway through which a given level of treatment delivery is reached. Two patients with similar final RDI may follow markedly different trajectories: one may begin treatment with preserved reserve and subsequently deteriorate following repeated toxicity, whereas another may enter treatment with substantial vulnerability and remain on an attenuated schedule. Baseline prognostic and geriatric frameworks provide important information on disease risk and patient fitness **[5,6]**, but they do not explicitly represent the feedback through which treatment reduces lymphoma burden while simultaneously affecting hematologic and functional reserve, which may in turn constrain subsequent treatment delivery.

This study motivates a dynamical rather than exclusively associative representation of treatment tolerance. We formulate first-line DLBCL treatment as a regime-switching dynamical system in which lymphoma burden, hematologic reserve, functional reserve, and cumulative treatment delivery evolve jointly over repeated treatment cycles. Regime-switching methods have been extensively used in financial engineering to represent systems whose behaviour changes between persistent states as underlying conditions evolve **[7]**. The concept is repurposed as a mathematical abstraction of transitions among preserved-reserve, compensated-vulnerability, and treatment-intolerance states. The analogy is methodological rather than clinical: the treatment states are not inferred financial regimes, and their boundaries are provisional modelling constructs rather than validated therapeutic thresholds. The resulting model combines nonlinear state evolution with state-dependent treatment delivery and explicitly propagates uncertainty in model parameters. Global sensitivity analysis and numerical robustness assessment are used to identify parameters governing simulated RDI and residual lymphoma burden **[8,9]**. The objective is not to establish a clinical dose-modification rule, but to provide a reproducible mechanistic proof-of-concept for investigating when evolving physiological reserve may alter treatment-delivery trajectories and for identifying quantities that should subsequently be calibrated and validated in longitudinal DLBCL cohorts.

Methods

Study Design and Modeling Scope

This study developed a theoretical, literature-informed regime-switching dynamical model to examine the interaction between treatment delivery and physiological reserve during first-line therapy for diffuse large B-cell lymphoma (DLBCL). The framework was designed as a mechanistic proof-of-concept rather than a patient-level prognostic or treatment-selection model. No individual patient data were used for parameter estimation, model fitting, or clinical validation. Clinical literature on relative dose intensity (RDI), treatment attenuation, comorbidity, and geriatric vulnerability informed the selection and interpretation of model variables **[1-6]**, whereas numerical parameter values and switching thresholds were specified as provisional computational quantities.

The primary modeling objective was to represent treatment tolerance as a state-dependent process in which lymphoma burden and physiological reserve evolve jointly over repeated treatment cycles. RDI was therefore treated as an emergent simulation output rather than a prespecified clinical treatment target. All simulated thresholds, regime assignments, and numerical outcomes should be interpreted as properties of the theoretical model rather than validated clinical cutoffs or patient-effect estimates.

State Variables Mapping

The model comprised four normalized time-dependent state variables:

$L(t) \in [0,1]$ (lymphoma burden)

$H(t) \in [0,1]$ (hematologic reserve)

$F(t) \in [0,1]$ (functional reserve)

$D(t)$ (cumulative delivered treatment intensity)

An additional normalized exogenous variable was defined as:

$C \in [0,1]$ (comorbidity pressure)

Table 1. State Variables and Observable Mapping

Symbol	Normalized state	Range	Potential clinical mapping
$L(t)$	Lymphoma burden	[0,1]	LDH, metabolic tumor volume, stage, response
$H(t)$	Hematologic reserve	[0,1]	Hemoglobin, neutrophils, platelets
$F(t)$	Functional reserve	[0,1]	ECOG, G8/frailty, albumin
$D(t)$	Cumulative delivered intensity	[0,1]	Delivered/planned intensity across cycles
C	Comorbidity pressure	[0,1]	Charlson index, CIRS-G, organ burden

Potential clinical mappings were prespecified for future calibration. Lymphoma burden may be informed by disease-related measures such as lactate dehydrogenase, metabolic tumor volume, stage, or treatment response; hematologic reserve by hemoglobin, neutrophils, and platelets; functional reserve by performance status, frailty or geriatric assessment, and albumin; and comorbidity pressure by established comorbidity indices. These mappings were conceptual and were not fitted to observed clinical measurements in the present study.

Dynamic Modelling

A Dynamic Treatment Reserve Index (DTRI) was defined to summarize the instantaneous balance between hematologic and functional reserve relative to lymphoma burden and comorbidity pressure:

$$DTRI(t) = \frac{\sqrt{H(t)F(t)}}{(1\ +\ 0.25L(t) +\ 0.25C)}$$

Treatment delivery was governed by the following piecewise multiplier u(t):

$$u(t) =\ 1.00,\quad DTRI(t) \geq\ 0.62$$
$$u(t) =\ 0.75,\quad 0.42\ \leq\ DTRI(t) <\ 0.62$$
$$u(t)\ =\ 0.55,\quad DTRI(t)\ <\ 0.42$$

The three states were interpreted operationally as preserved reserve, compensated vulnerability, and treatment intolerance, respectively. The regime-switching construction was conceptually motivated by state-dependent systems in which system behavior changes when an underlying state crosses a specified boundary [7]. The thresholds 0.62 and 0.42 and corresponding treatment multipliers were modeling assumptions rather than empirically derived clinical decision thresholds.

$$\frac{dL}{dt}\ =\ gL\left(1-\frac{L}{K}\right)-\ eu(R)L$$
$$\frac{dH}{dt}\ =\ rH(1-H)F\ -\ tH\,u(R)H\ -\ dHLH\ -\ cHCH$$
$$\frac{dF}{dt}\ =\ rF(1-F)H\ -\ tFu(R)F\ -\ dFLF\ -\ cFCF$$
$$\frac{dD}{dt} = \frac{u(R)}{6}$$
$$DTRI(t) = \frac{\sqrt{H(t)F(t)}}{(1\ +\ 0.25L(t) +\ 0.25C)}$$

The treatment multiplier $u(R)$ equals 1.00 when $DTRI \geq 0.62$, 0.75 when $0.42 \leq DTRI < 0.62$, and 0.55 when $DTRI < 0.42$. These operational regimes represent preserved reserve, compensated vulnerability, and treatment intolerance. They are modeling devices rather than proposed bedside thresholds. $D(6)$ is interpreted as modeled RDI over six cycle units. The vector field is locally Lipschitz within each regime. Boundary evaluation establishes positive invariance of $[0, K]x[0,1]x[0,1]$. Piecewise uniqueness is retained across right-continuous switching surfaces, and boundedness gives global continuation over the treatment horizon. Regime-specific equilibria satisfy the algebraic system $f(x) = 0$, local stability is determined by Jacobian eigenvalues. Numerical continuation in baseline reserves, toxicity, treatment effect, and comorbidity pressure identify changes in regime occupancy and final basin membership.

Parameterization and Numerical Methods

Reference simulations used $L0 = 0.75, H0 = 0.72, F0 = 0.68, C = 0.35$, lymphoma growth $g = 0.22$, treatment effect $e = 0.62$, hematologic recovery $rH = 0.18$, functional recovery $rF = 0.15$, hematologic toxicity $tH = 0.17$, and functional toxicity $tF = 0.15$. Fourth-order Runge-Kutta integration covered six cycle units with step 0.01. Robustness analyses comprised five clinical scenarios, a 41x41 reserve surface, threshold continuation, 500-set Latin hypercube sampling, PRCC, stochastic parameter perturbation, and repeated integration at fourfold coarser resolution.

Reproducible Scenario Results

Scenario	Initial RI	RDI	Final L	Final H	Final F	Final RI	Intolerant time
High reserve	0.687	0.785	0.103	0.476	0.484	0.446	0.0%
Reference	0.549	0.674	0.149	0.450	0.443	0.397	38.3%
Borderline reserve	0.458	0.590	0.193	0.419	0.412	0.362	79.9%
Low reserve	0.423	0.553	0.211	0.399	0.395	0.340	98.3%
Severely constrained	0.340	0.550	0.213	0.343	0.336	0.285	100.0%

The reference trajectory delivered modeled RDI 0.674 and reduced lymphoma burden by 80.1% over six cycle units. High reserve increased RDI by 16.6% and reduced final lymphoma burden by 30.8% relative to reference. Low reserve reduced RDI to 0.553, the severely constrained trajectory remained treatment-intolerant for 100.0% of simulated time. Along $H0 = F0$, the illustrative $RDI \geq 0.70$ boundary occurred near 0.745. These are deterministic model contrasts, not patient-effect estimates.

Global Sensitivity and Uncertainty

Parameter	PRCC with RDI	PRCC with final burden
$F0$	0.872	-0.692
$H0$	0.859	-0.708
C	-0.769	0.596
tf	-0.694	0.518
th	-0.653	0.476
rh	0.509	-0.361
rf	0.478	-0.301
dh	-0.375	0.262
e	0.339	-0.949
df	-0.284	0.176
g	-0.178	0.836

Across 500 uncertainty sets, median modeled RDI was 0.651 with a 5th-95th percentile interval of 0.567-0.753. Final burden had a median of 0.165 and interval 0.081-0.288. Solver coarsening changed reference outputs by at most 0.000167 in normalized units.

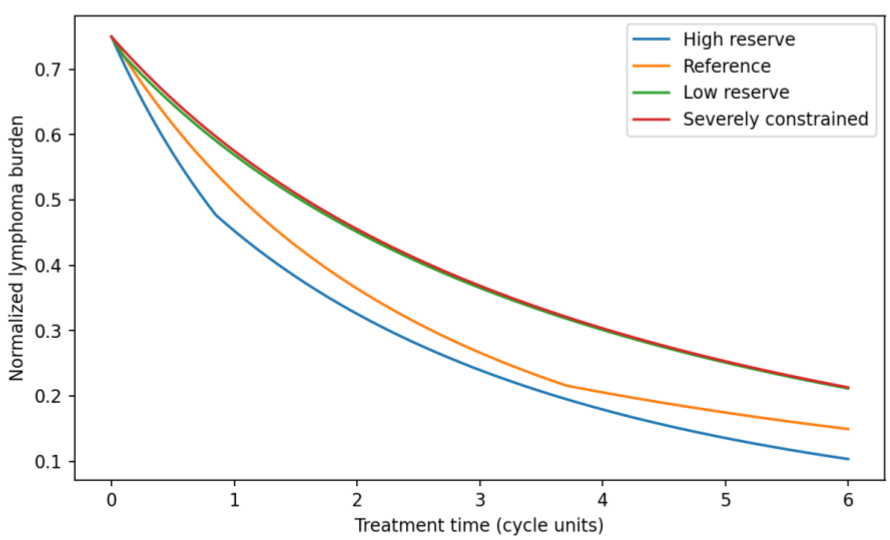

Figure 1. Lymphoma-burden trajectories.

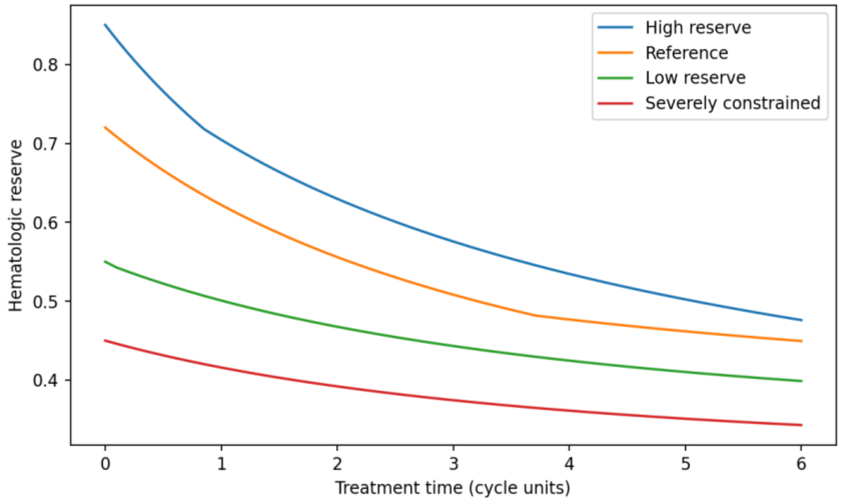

Figure 2. Hematologic-reserve trajectories.

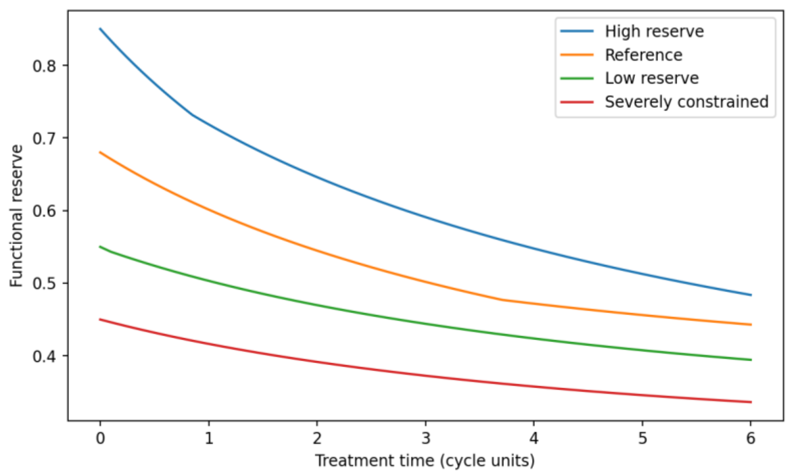

Figure 3. Functional-reserve trajectories.

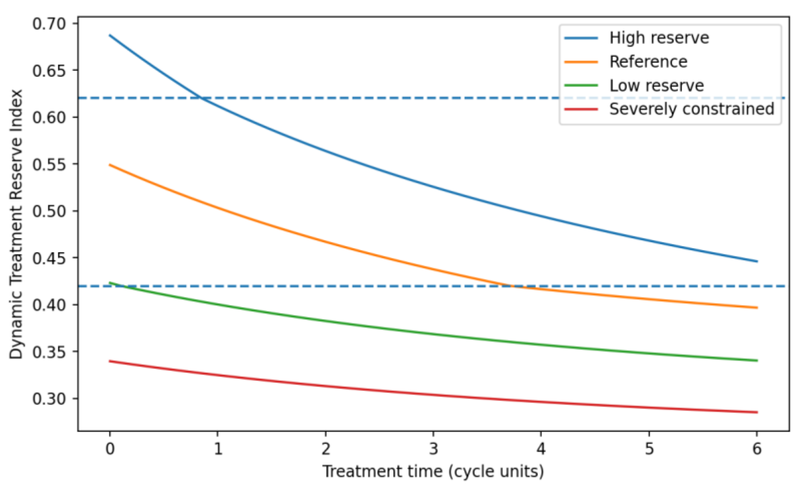

Figure 4. Dynamic Treatment Reserve Index and regime boundaries.

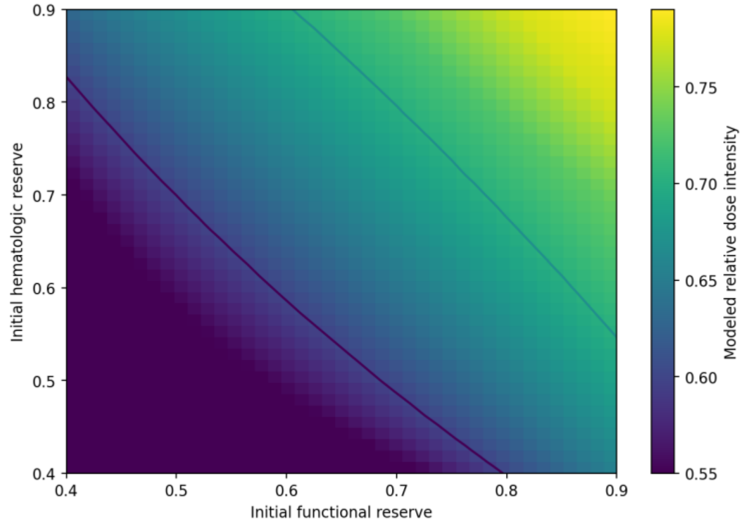

Figure 5. Dynamic risk surface linking initial reserves to modeled RDI.

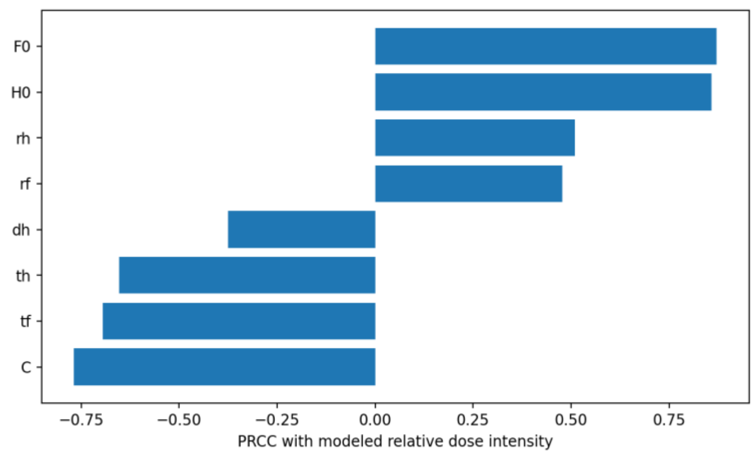

Figure 6. Global sensitivity of modeled RDI.

Discussion

The model separates lymphoma control from the capacity to keep delivering treatment. Patients with the same baseline burden can diverge because hematologic recovery, functional recovery, toxicity, and comorbidity alter future regime occupancy. The dynamic risk surface does not define a universal RDI target. It identifies combinations of reserve and toxicity that keep treatment delivery within a mathematically recoverable region. This state-dependent information is not available from a single baseline regression coefficient. The model aggregates several biological processes and does not yet include cycle-specific laboratory nadirs, infection, organ-specific toxicity, treatment cessation, competing death, or progression as separate events. DTRI thresholds and parameters are provisional. The framework has not been fitted to patients and cannot support treatment recommendations. Sensitivity, specificity, AUC, calibration slope, Brier score, and decision utility are inappropriate until a patient-level cohort, observation model, and prespecified endpoint are available.

Conclusion

This mechanistic proof-of-concept establishes a mathematically testable structure for locating treatment-tolerance windows during first-line DLBCL therapy. Its intended uses are selecting serial reserve variables, refining dose-delivery hypotheses, designing retrospective or prospective protocols, and supporting collaboration between mathematical modelers, hematologists, geriatricians, and clinical pharmacists. The potential benefit is earlier recognition of reserve collapse before repeated interruption becomes entrenched. Its research advantages are explicit state dependence, transparent assumptions, reproducible counterfactual scenarios, formal uncertainty analysis, and clear separation between model-generated hypotheses and validated treatment rules.

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